

MP307 Practical 2019/2020: Continuous Population Models

This practical uses the Maple worksheet `continuous.mw` that can be downloaded from the MP307 Blackboard page.

Notice

Solutions to **all** questions marked with (*) has to be shown (and explained) to the instructor at the practicals in order to get 4% that count towards the overall mark.

1. **Verhulst Logistic Model.** Consider the logistic population model

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$$

for $r, K > 0$ as a model of the US population from 1790 to 1990. The data for this are given in the worksheet `cont.mw`.

2. (*) **Competitive Species.** Consider two species with population sizes P_1, P_2 with growth rates r_1, r_2 and limiting population sizes of K_1, K_2 where

$$\begin{aligned}\frac{dP_1}{dt} &= r_1 P_1 \left(1 - \frac{P_1 + P_2}{K_1}\right), \\ \frac{dP_2}{dt} &= r_2 P_2 \left(1 - \frac{P_1 + P_2}{K_2}\right).\end{aligned}$$

Analyse the behaviour of N_1, N_2 in the following cases, by plotting P_1, P_2 vs t and P_1 vs P_2 :

- (a) $r_1 = 1/10, r_2 = 1/10, K_1 = 100, K_2 = 50$ with $P_1(0) = 10$ and $P_2(0) = 15$.
 - (b) $r_1 = 1/10, r_2 = 1/10, K_1 = 100, K_2 = 50$ with $P_1(0) = 130$ and $P_2(0) = 200$.
 - (c) $r_1 = 1/10, r_2 = 1/100, K_1 = 40, K_2 = 50$ with $P_1(0) = 130$ and $P_2(0) = 20$.
 - (d) $r_1 = 1/10, r_2 = 1/100, K_1 = 50, K_2 = 60$ with $P_1(0) = 15$ and $P_2(0) = 10$.
3. (*) **Lotka-Volterra Predator/Prey System.** Consider a prey species with population size x and a predator species with population size y where

$$\begin{aligned}\frac{dx}{dt} &= x(a_1 - b_1 y), \\ \frac{dy}{dt} &= y(-a_2 + b_2 x),\end{aligned}$$

with $a_1, a_2, b_1, b_2 > 0$. Analyse the behaviour of the system for $a_1 = 3, a_2 = 5/2, b_1 = 2, b_2 = 1$ by plotting x, y vs t , and x vs y in the following cases:

- (a) $x(0) = 1$ and $y(0) = 1$.
- (b) $x(0) = 0.1 + 5/2$ and $y(0) = 0.1 + 3/2$. What behaviour do you observe?
- (c) $x(0) = 1$ and $y(0) = 5$.